

Fundamentals of regression and classification.

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Objectives

1. Train regression and classification models.
2. Evaluate the performance of the trained models.
3. Understand the assumptions and the formulation of some of the basic regression and classification models and relate them to the obtained results.

Introduction

In this practical session, we will train and evaluate simple regression and classification models in several public datasets: the Concrete Compression Dataset, the Superconductivity Dataset and the CIFAR-10 Database.

Evaluation

You need to **submit it in Moodle before the next lab session** (check the exact deadline in moodle). To do this:

1. make a copy of the colab notebooks,
2. write the code and report in the same notebook,
3. download all the colab documents as .ipynb and upload them to moodle.

The following will be evaluated:

- Correctness and extent of the work.
- Correctness in the use of technical terms in the explanations.
- Analysis and discussion of the results.
- Organization and cleanliness of the code.

1. Regression in the Concrete Compression Dataset

The goal of this first part of the practical is to train and evaluate a regression model. We will use the Concrete Compression Dataset. The dataset was first presented in: Modeling of strength of high-performance concrete using artificial neural networks By I. Yeh. 1998 Published in Cement and Concrete Research, Vol. 28, No. 12. Find an overview here <https://archive-beta.ics.uci.edu/dataset/165/concrete+compressive+strength>

Start your work from the following [notebook](#), which loads the data.

1.1 Split the data

First of all, after you have your data, split it into the training, validation and test sets. Make sure you do not look at the test set until you report your final metrics.

1.2 Explore the data

Report first the size of the problem (how many samples are in the dataset? How many dimensions does your data have?).

Plot and/or analyze in more depth the relation of each dimension with the output. For example, for each input variable and the output, 1) compute the correlation (you can use the `numpy` function `corrcoef`) and 2) plot the relation between the two variables in a 2-d graph (you can use the `pyplot` function `scatter`).

1.3 Train several promising models, decide on the best one and report expected performance

Train the regression models we have seen in class (linear regression, polynomial models, ridge/Huber regression, RANSAC...) and others if you want to. If training is slow, consider doing quick explorations reducing the dataset size or the data dimensionality. Report several metrics and analyze their values. Extract conclusions and explain the results.

1.4 High dimensional data

You can train similar regression models with the Superconductivity dataset using this [notebook](#). Note that, in this case we have 81 features. In this case, it might be interesting to reduce the dimensionality of the problem with methods such as PCA or reduce the number of features with methods such as LASSO. Remember that data is the best regularizer, thus it is interesting to compare different models with different data sizes. You can select the training data size with the `train_test_split` function using the `train_size` parameter https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html

You can read about the dataset here

https://www.openml.org/search?type=data&sort=version&status=any&order=asc&exact_name=superconductivity&id=44964

2 Classification in the CIFAR-10 dataset

The goal of the second part of the practical is to train and evaluate classification models in the CIFAR10 dataset, a toy dataset for visual classification. Find an overview of the dataset in <https://www.cs.toronto.edu/~kriz/cifar.html> Start your work from the following [notebook](#), which loads the data.

2.1 Split the data, explore it and train several promising models

As in the previous case, split your data into training, validation and test sets. Show the images and display the size of the dataset. Finally, train and evaluate several classification models from the ones we learned in class. If training is slow, remember that you can use PCA to reduce the dimensionality of the data or you can try things out with smaller versions of the full dataset.

2.2 Transform the data

As you might imagine, finding appropriate representation of the visual data is complex. And as I told you in class, finding appropriate features is crucial for learning methods to work. For you to understand the importance of using good features, we will use here the so-called GIST descriptors, for which you have code to extract them in the notebook of the practical. Originally proposed in *Aude Oliva and Antonio Torralba. Modeling the shape of the scene: a holistic representation of the spatial envelope. International Journal of Computer Vision, Vol. 42(3) (2001) 145-175*. It basically convolves the image with [Gabor filters](#) at different orientations and scales, and averaging their responses in a grid.

Compute the GIST descriptors of all the CIFAR-10 images, and use them as the feature vectors to train again your models. Select reasonable values for the hyperparameters of your models and report the performance of your classifiers in the test set.

2.3 Show examples of the classification results

Once your classifier is trained, show several examples for which your classifier predicted the correct label, and some others for which your classifier predicted the wrong one.